

Ricardo Castro Figueiredo

Full-Stack Developer

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[LinkedIn](#) • [GitHub](#) • [Portfolio](#)

Experience

Skill & Reach

Software Developer – Intern

Porto, Portugal

Oct. 2025 – Dec. 2025

- Built an end-to-end **OSINT** news ingestion pipeline aggregating **RSS feeds** and scraped web sources using **Python**, processing 1000+ news articles daily.
- Applied **LLM**-based enrichment for article classification and summarization, with heuristic fallbacks.
- Implemented semantic deduplication and related-article linking using vector embeddings and **Elasticsearch**.
- Developed a **FastAPI** backend and **React** control center for article management and automated email reports.

Critical Manufacturing

Software Developer – Intern

Maia, Portugal

July 2024 – Sept. 2024

- Designed and deployed a **full-stack** performance appraisal platform to automate internal employee evaluations.
- Developed a modular frontend using component-based **vanilla JavaScript** and HTML/CSS.
- Engineered a **REST** backend with **Node.js** and **Express.js**, implementing custom Middleware for JWT-based authentication and Role-Based Access Control (RBAC).
- Designed a **MySQL** relational schema to manage feedback workflows between employees and appraisers.

Projects

- **Shinsu Duel: a Tower of God CCG** *Node.js, Socket.IO, Express.js*
 - Real-time PvP/PvE card game with synced state management.
 - Component-based UI with drag-and-drop interactions.
- **Kotlin to MIPS Compiler** *Haskell, Alex/Happy, MIPS*
 - Developed compiler with lexer/parser (Alex/Happy), semantic checks, and MIPS codegen for Kotlin constructs.
 - Optimized via graph coloring register allocation and liveness analysis.
- **CART Class Imbalance** *Python, scikit-learn, SMOTE*
 - Improved Recall score by 18% with weighted Gini impurity for minority classes.
 - Validated via stratified sampling and ROC-AUC metrics across several datasets.
- **Roomba Battle Bot** *Java, Robocode, OOP*
 - Predictive targeting using trigonometry and adaptive bullet power.
 - Energy-efficient evasion via collision timeout tracking + zigzag movement.
- **Cryptography Benchmark** *Python, Cryptography, Matplotlib*
 - Automated performance analysis of AES/RSA/SHA-256.
 - Visualized encryption-decryption trends with statistical averaging.

Education

Bachelor in Computer Science

Feb. 2026

University of Porto

- **Relevant Coursework:** Algorithms & Data Structures, AI/ML, Computer Systems, Databases & Big Data, Security & Cryptography, Software Architecture, Mathematics, Programming Paradigms, Theory of Computation, Human-Machine Interaction

Skills

- **Languages:** Python, Java, C++, Haskell, SQL, JavaScript, TypeScript, HTML/CSS, Bash/Shell, LaTeX
- **Tools & Frameworks:** Node.js, Express.js, FastAPI, REST APIs, MySQL, MongoDB, Elasticsearch, React, Git, LLMs, Pandas, NumPy, TCP/IP, AMPL, Jupyter
- **Platforms:** AWS, GCP, Linux/Unix, Docker

Certifications

- English C1 Advanced (CAE), Cambridge Assessment English

June 2021